

SCTS - DATABASE SYSTEMS SYLLABUS

Course Code: CS-402 | Core Module | Credits: 4

Course Overview: Relational database design, query optimization, and transaction handling.

Unit 1: Entity-Relationship Models, Relational Algebra, Relational Calculus.

Unit 2: Advanced SQL Query Writing, Joins, Triggers, Views, and Indices.

Unit 3: Normalization Forms - 1NF, 2NF, 3NF, BCNF, 4NF, Lossless Decomposition.

Unit 4: Transaction Processing, ACID Properties, Concurrency Control, Lock Manager.

Reference Books:

1. Database System Concepts - Abraham Silberschatz, Henry F. Korth
2. Fundamentals of Database Systems - Ramez Elmasri